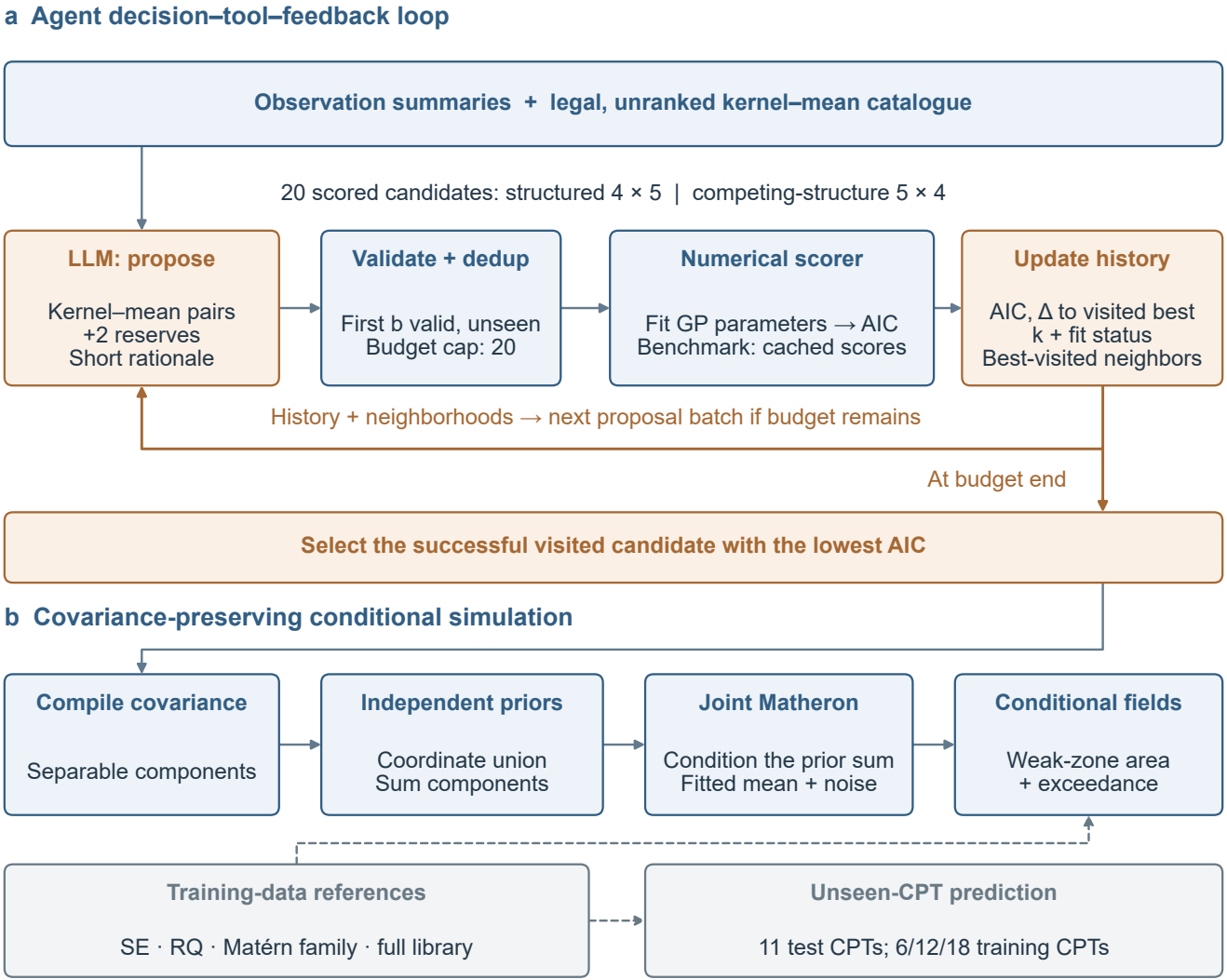
a Agent decision–tool–feedback loop
Observation summaries + legal, unranked kernel–mean catalogue
20 scored candidates: structured 4 × 5 | competing-structure 5 × 4
LLM: propose
Kernel–mean pairs
+2 reserves
Short rationale
Validate + dedup
First b valid, unseen
Budget cap: 20
Numerical scorer
Fit GP parameters → AIC
Benchmark: cached scores
Update history
AIC, Δ to visited best
k + fit status
Best-visited neighbors
History + neighborhoods → next proposal batch if budget remains
At budget end
Select the successful visited candidate with the lowest AIC
b Covariance-preserving conditional simulation
Compile covariance
Separable components
Independent priors
Coordinate union
Sum components
Joint Matheron
Condition the prior sum
Fitted mean + noise
Conditional fields
Weak-zone area
+ exceedance
Training-data references
SE · RQ · Matérn family · full library
Unseen-CPT prediction
11 test CPTs; 6/12/18 training CPTs